Primary Six Mathematics — Finding unknown prime factors

Name: _____ Date: _____

Learn and understand

Use factor trees and shared prime factors. For positive whole numbers a and b, HCF(a,b) × LCM(a,b) = a × b. After calculating an unknown number, always verify BOTH its HCF and LCM with the known number.

Worked examples

Example 1: Two numbers are 12 and x. Their HCF is 6 and LCM 60. Find x.

Working: 6×60=12×x; x=30.

Example 2: If $36=2^2\times3^2$, which prime factor is missing in 2×2×3×__?

Working: 3.

Practice independently

Show your working and units.

1. Find x if HCF=3, LCM=30, a=6.

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_____

2. Find x if HCF=5, LCM=60, a=15.

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3. Missing factor: 30=2×3×__.

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_____

4. Missing factor: 84=2×2×3×__.

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5. HCF of 18 and 27.

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6. LCM of 18 and 27.

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7. HCF×LCM for 8 and 12.

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8. Find b if HCF=4, LCM=40, a=8.

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Check your understanding

Explain the rule you used: _____